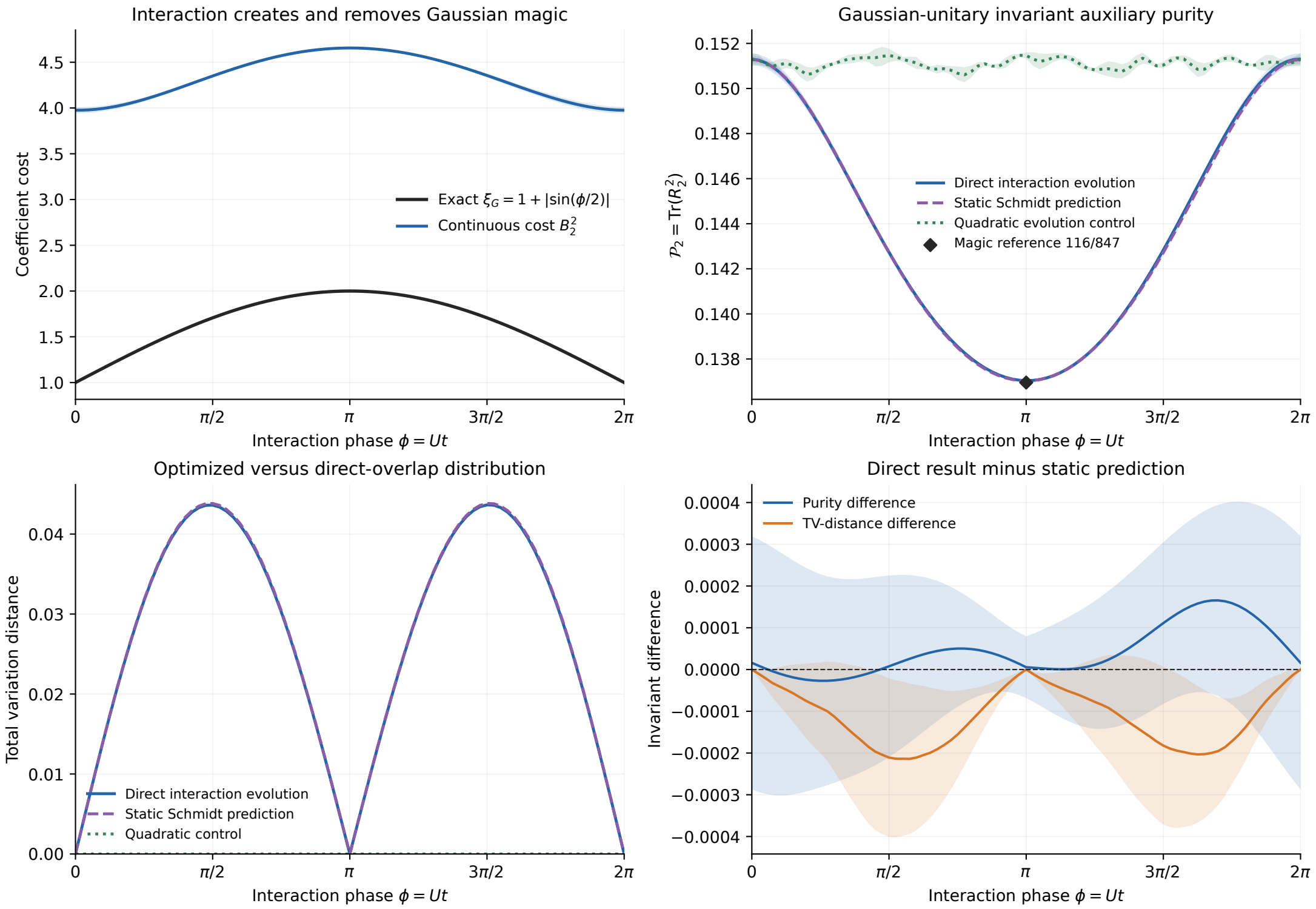
Quartic interaction: direct evolution and static prediction
Interaction creates and removes Gaussian magic
Coefficient cost
Exact $\xi_G = 1 + |\sin(\phi/2)|$
Continuous cost B_2^2
Interaction phase $\phi = Ut$
Gaussian-unitary invariant auxiliary purity
$\mathcal{P}_2 = \mathrm{Tr}(R_2^2)$
Direct interaction evolution
Static Schmidt prediction
Quadratic evolution control
Magic reference 116/847
Interaction phase $\phi = Ut$
Optimized versus direct-overlap distribution
Total variation distance
Direct interaction evolution
Static Schmidt prediction
Quadratic control
Interaction phase $\phi = Ut$
Direct result minus static prediction
Invariant difference
Purity difference
TV-distance difference
Interaction phase $\phi = Ut$
Bands: +/-1 standard error. Direct and static predictions use independent observation streams.